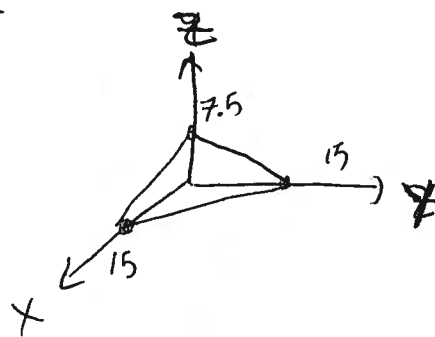


# 11.8 Surfaces in 3-space

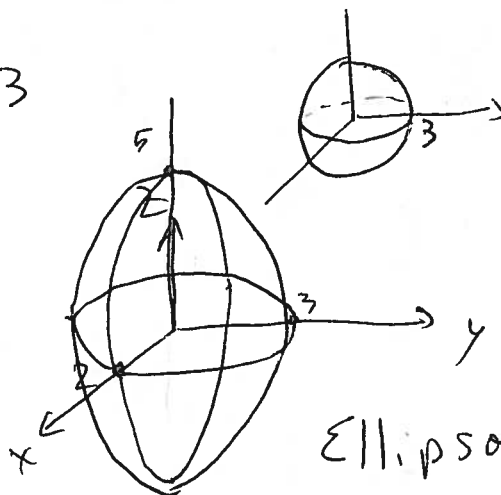
$\Sigma_{x1}$   $Ax + By + Cz = D$  a plane

eg  $x + y + 2z = 15$



$\Sigma_{x2}$   $x^2 + y^2 + z^2 = 9$

Sphere of radius 3

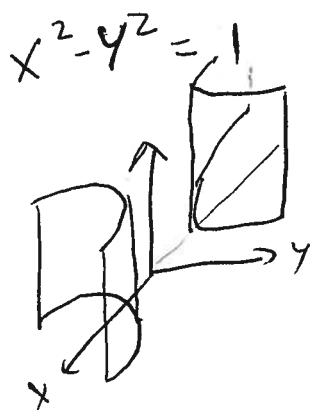
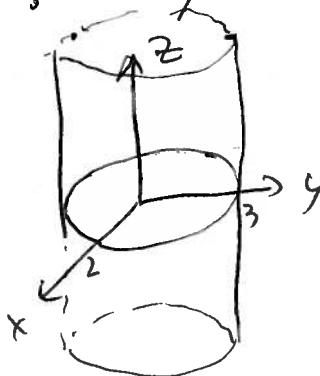


$\Sigma_{x3}$   $\frac{x^2}{4} + \frac{y^2}{9} + \frac{z^2}{25} = 1$

Intersections with coordinate planes are ellipses

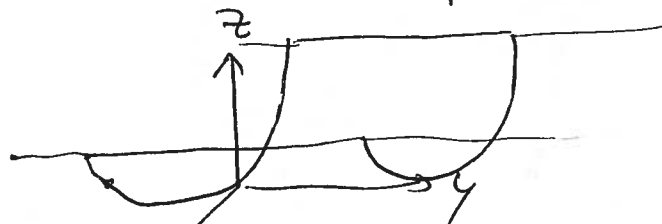
Ellipsoid  
 $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$

$\Sigma_{x4}$   $\frac{x^2}{4} + \frac{y^2}{9} = 1$  no  $z$ ! cylinder parallel to  $z$ -axis

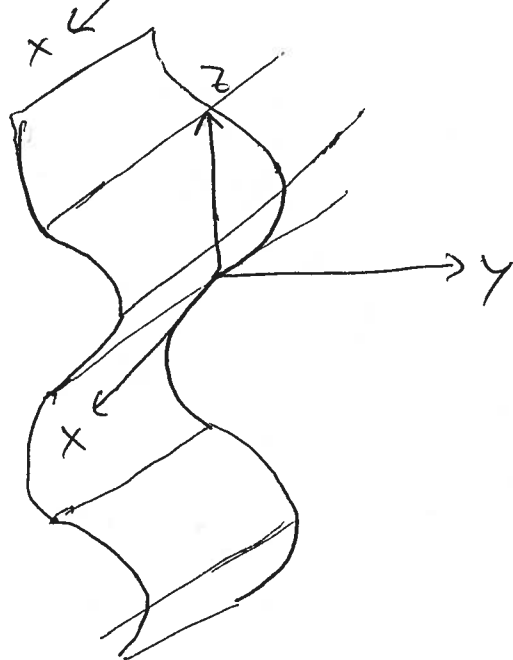


$$x^2 - z = 0$$

no  $y$  so parallel to  $y$ -axis



$y = \sin z$   
parallel to  
 $x$ -axis

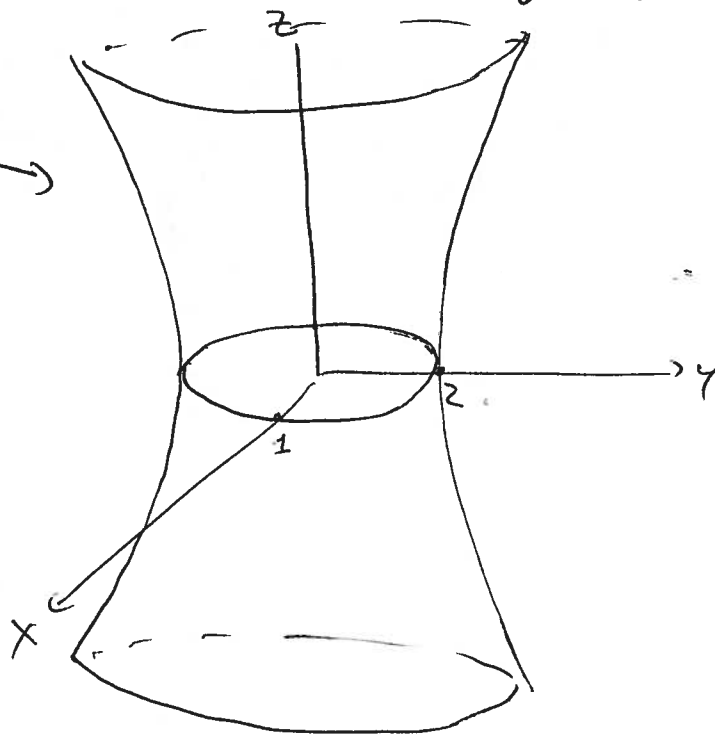
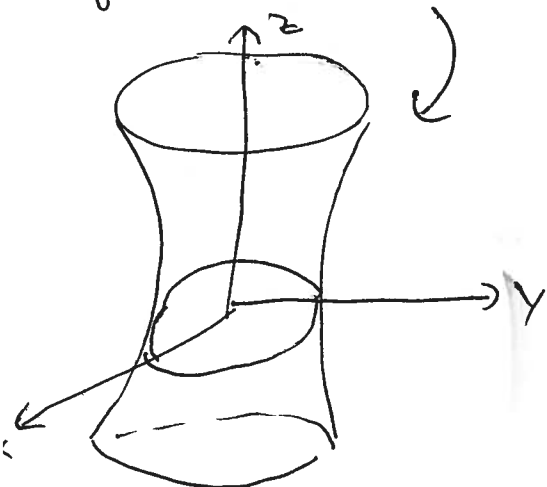


Ex 5 One sheet hyperboloid

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$$

eg  $x^2 + \frac{y^2}{4} - \frac{z^2}{9} = 1$

eg  $x^2 + y^2 = 1 + z^2$



## Hyperboloid of two sheets

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$$

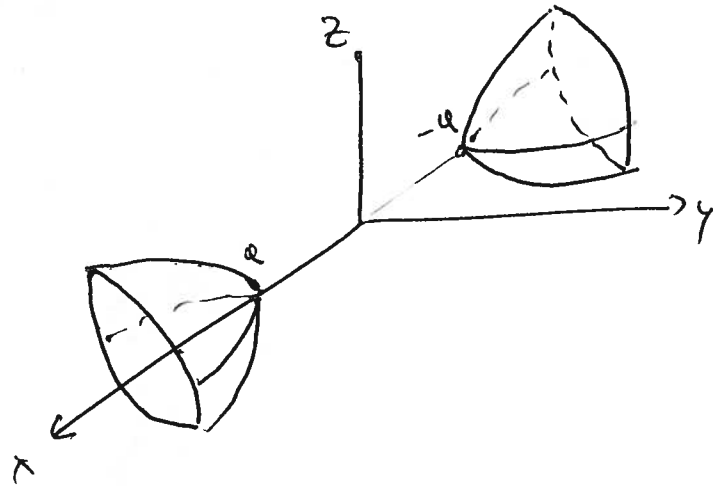
eg  $x^2 - y^2 - z^2 = 1$

or  $-x^2 + 2y^2 - z^2 = 1$

↳  $x^2 + z^2 = 2y^2 - 1$

for  $y > \frac{1}{\sqrt{2}}$  &  $y < -\frac{1}{\sqrt{2}}$

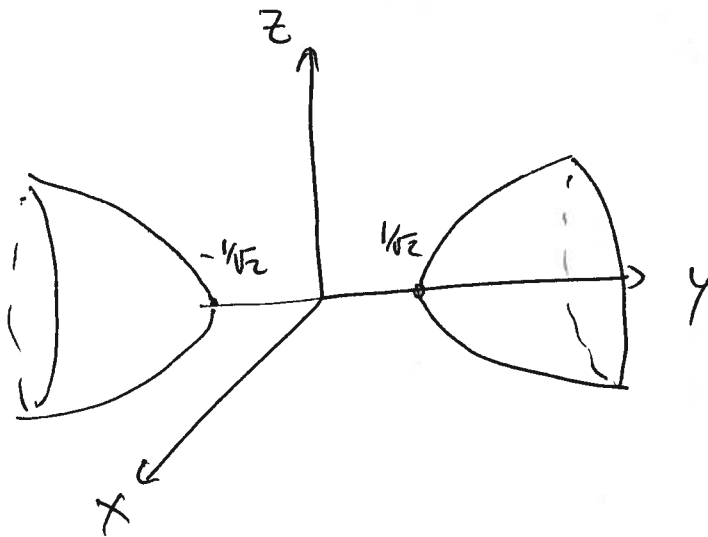
it's a circle of  
radius  $\sqrt{2y^2 - 1}$



If  $x = c$  then

$$2y^2 - z^2 = 1 + x^2 = 1 + c^2$$

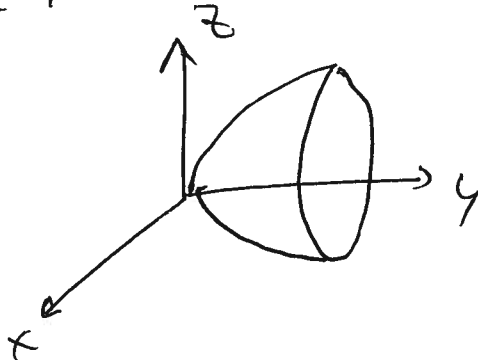
hyperbola



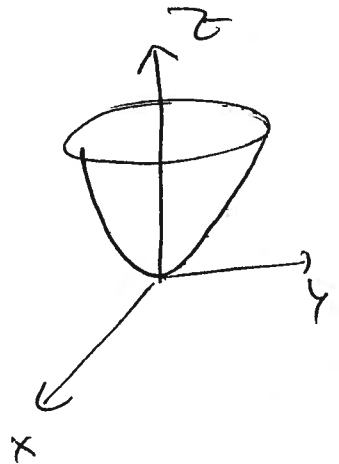
## Elliptic Paraboloid

for each  $z = c > 0$   
we get an ellipse

eg  $y = x^2 + z^2$

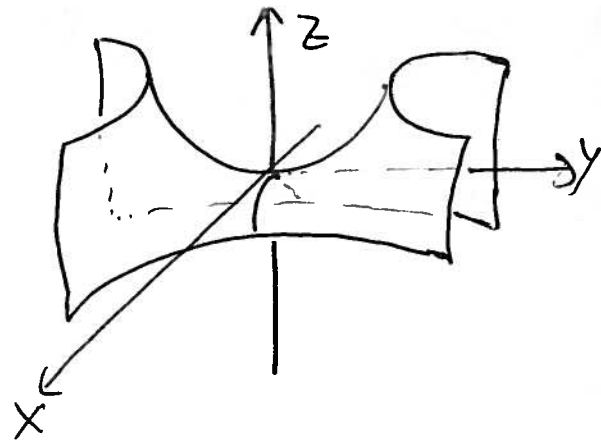


$$z = \frac{x^2}{a^2} + \frac{y^2}{b^2}$$



# Hyperbolic Paraboloid or saddle

$$z = \frac{y^2}{b^2} - \frac{x^2}{a^2}$$



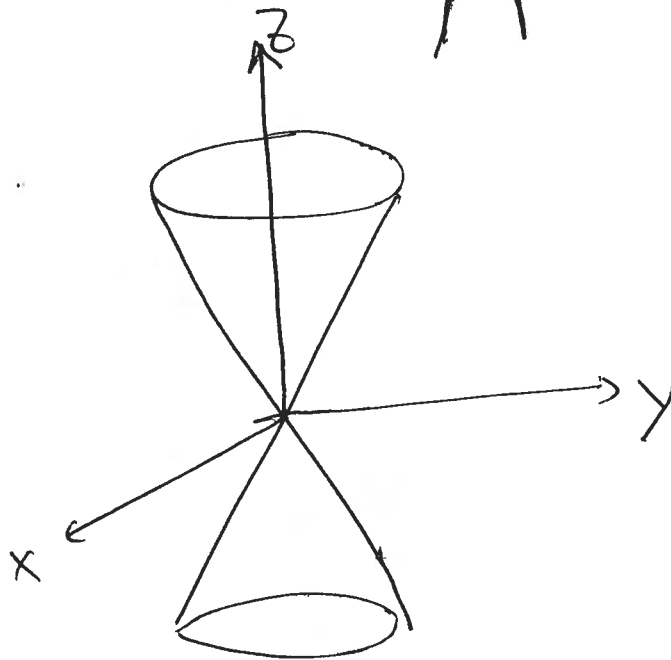
for each  $z=c$  we get  
a hyperbola.

for each  $x=c$  we get parabola  
" "  $y=c$  " " " "

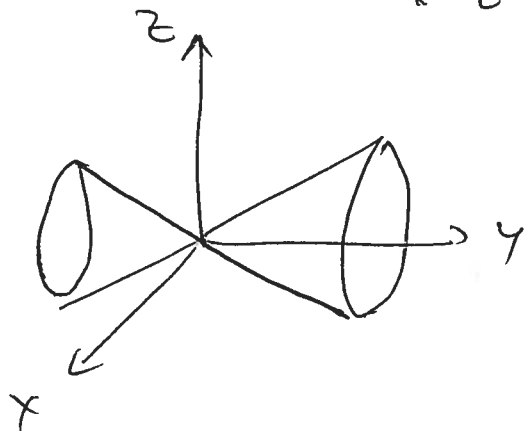


# Elliptic cone

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 0$$



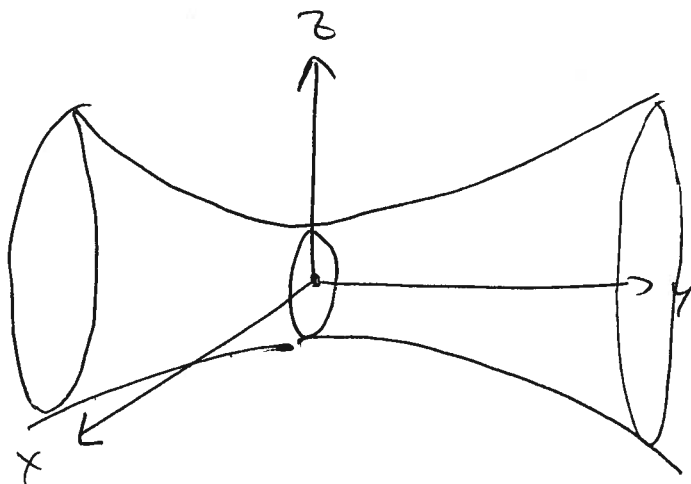
Ex  $x^2 - y^2 + z^2 = 0$   
 $\rightarrow x^2 + z^2 = y^2$



$\Sigma x$ 

$$-x^2 + y^2 + z^2 = 1$$

$$\rightarrow y^2 + z^2 = 1 + x^2$$

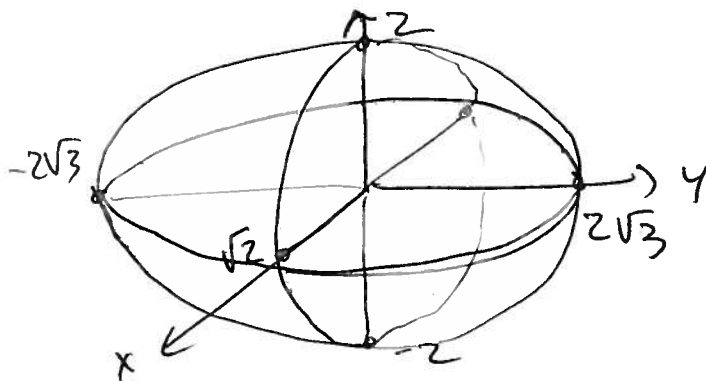


one sheet  
hyperboloid

 $\Sigma x$ 

$$2x^2 + \frac{y^2}{3} + z^2 = 4$$

$$\Rightarrow \frac{x^2}{2} + \frac{y^2}{12} + \frac{z^2}{4} = 1$$

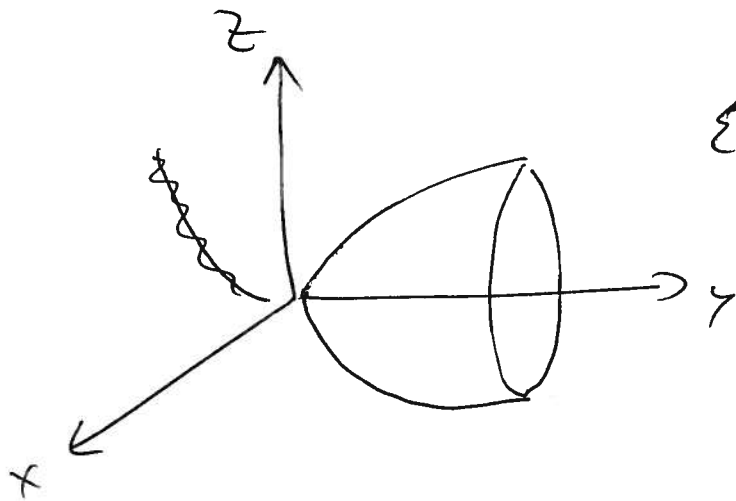


ellipsoid

 $\Sigma x$ 

$$x^2 - 2y + 3z^2 = 0$$

$$\rightarrow y = \frac{x^2}{2} + \frac{3}{2}z^2$$



elliptic  
paraboloid

Ex graph the following in  $\mathbb{R}^3$

1.  $4x^2 + 36y^2 = 144$

2.  $3x + 2z = 10$

3.  $x^2 + y^2 - 8x + 4y + 13 = 0$

4.  $4x^2 + 9y^2 + 49z^2 = 1764$

5.  $4x^2 + 16y^2 - 32z = 0$

6.  $y = e^z$

7.  $x^2 - z^2 + y = 0$

8.  $-x^2 + y^2 + z^2 = 1$

9.  $-x^2 - y^2 + z^2 = 1$